ASSIGNMENT 2

Arduino and Microcontroller Concepts

B.Tech. 7th Semester • 2-mark exam-ready questions and answers

1 Write an Arduino sketch to blink an LED 20 times with a delay of 250 ms.

ANSWER

```
void setup() {
  pinMode(13, OUTPUT);
}
void loop() {
  for (int i = 0; i < 20; i++) {
    digitalWrite(13, HIGH); delay(250);
    digitalWrite(13, LOW);  delay(250);
  }
  while (1);
}
```

The LED turns ON and OFF 20 times, with a 250 ms delay in each state.

2 Compare ultrasonic and optical ToF sensors. Choose sensors for detecting object movement and human presence.

ANSWER

- **Ultrasonic sensor:** Uses sound waves to measure distance; affected by soft surfaces, temperature and environmental noise.
- **ToF sensor:** Uses light pulses to measure distance accurately; affected by strong sunlight and surface reflectivity.
- **Tilt sensor:** Detects changes in an object's orientation.
- **PIR sensor:** Detects people through changes in infrared radiation.

A tilt sensor is suitable for detecting object movement, while a PIR sensor is suitable for detecting nearby people.

3 Convert a sensor reading (0–1023) into temperature (0–100 °C) using map().

ANSWER

```
void setup() { Serial.begin(9600); }
void loop() {
  int sensor = analogRead(A0);
  int temp = map(sensor, 0, 1023, 0, 100);
  Serial.println(temp);
  delay(500);
}
```

The map() function converts the sensor range 0–1023 into 0–100 °C.

4 Differentiate between delay(), delayMicroseconds() and millis().

ANSWER

- **delay()** pauses the program for a specified time in milliseconds.
- **delayMicroseconds()** pauses it for a specified time in microseconds.
- **millis()** returns the milliseconds elapsed since the program started.

millis() tracks elapsed time without blocking. Its value eventually overflows, so use unsigned subtraction for elapsed-time calculations.

5 Explain digitalRead(), analogRead() and calculate the analog value for 3.5 V.

ANSWER

- Button pressed = HIGH (1); released = LOW (0).
- A 10 kΩ pull-down resistor keeps the input LOW when released and prevents a floating input.
- digitalRead() reads HIGH/LOW; analogRead() reads an analog voltage level.

Calculation: ADC = (3.5 / 5) × 1023 ≈ 716.

6 Convert an analog reading into voltage and display a warning above 3 V.

ANSWER

```
void setup() { Serial.begin(9600); }
void loop() {
  int value = analogRead(A0);
  float voltage = value * (5.0 / 1023.0);
  Serial.println(voltage);
  if (voltage > 3.0) Serial.println("WARNING: High Voltage");
  delay(500);
}
```

Arduino Uno uses a 10-bit ADC (0–1023). The assumed reference voltage is 5 V.

7 State the Serial Monitor output as digital pin 10 changes between LOW and HIGH.

ANSWER

```
int pin10;
void setup() { pinMode(10, INPUT); Serial.begin(9600); }
void loop() {
  int ip = digitalRead(10);
  if (ip > 0) Serial.println("HIGH");
  else Serial.println("LOW");
}
```

Input LOW (0) displays LOW; input HIGH (1) displays HIGH. Each reading appears on a new line.

8 Control LED brightness through Serial Monitor. Differentiate print() and println().

ANSWER

```
void setup() { pinMode(9, OUTPUT); Serial.begin(9600); }
void loop() {
  if (Serial.available() > 0) {
    int brightness = Serial.parseInt();
    brightness = constrain(brightness, 0, 255);
    analogWrite(9, brightness);
  }
}
```

`Serial.print("A"); Serial.print("B");` displays AB on one line. println() prints each value on a new line.

9 Explain pinMode() with an example. Give two string functions.

ANSWER

```
pinMode(13, OUTPUT);
digitalWrite(13, HIGH);
```

pinMode() configures a pin as INPUT, OUTPUT or INPUT_PULLUP.

Example string: `String name = "Arduino";`

- **length()** returns the number of characters.
- **toUpperCase()** converts the string to uppercase.

10 Explain overflow, its handling and commonly used Arduino data types.

ANSWER

Overflow occurs when a variable exceeds its storage range. It can be handled by choosing a larger data type or checking limits before calculations.

Example: `long sum = 100000L + 200000L;`

- **int** – integer
- **long** – larger integer
- **float** – decimal
- **char** – character
- **bool** – true/false
- **byte** – 0 to 255

11 Turn ON an LED when a potentiometer reading exceeds 500.

ANSWER

```
void setup() { pinMode(13, OUTPUT); }
void loop() {
  int value = analogRead(A0);
  if (value > 500) digitalWrite(13, HIGH);
  else digitalWrite(13, LOW);
}
```

The LED turns ON when the analog reading is greater than 500; otherwise, it turns OFF.

12 Measure distance using HC-SR04. Calculate distance for an echo duration of 2000 µs.

ANSWER

```
void setup() {
  pinMode(7, OUTPUT); pinMode(8, INPUT);
  Serial.begin(9600);
}
void loop() {
  digitalWrite(7, LOW);  delayMicroseconds(2);
  digitalWrite(7, HIGH); delayMicroseconds(10);
  digitalWrite(7, LOW);
  long duration = pulseIn(8, HIGH);
  float distance = (duration * 0.0343) / 2;
  Serial.println(distance); delay(500);
}
```

Distance = (2000 × 0.0343) / 2 = 34.3 cm.

13 Identify the output: a = 22, b = 13, c = (a & b) | 2.

ANSWER

22 = 10110; 13 = 01101.

22 AND 13 = 00100 = 4.

4 OR 2 = 6.

Output: 6

14 Control two appliances using Serial Monitor commands L1ON, L1OFF, L2ON and L2OFF.

ANSWER

```
void setup() {
  pinMode(8, OUTPUT); pinMode(9, OUTPUT);
  Serial.begin(9600);
}
void loop() {
  if (Serial.available() > 0) {
    String cmd = Serial.readStringUntil('\n'); cmd.trim();
    if (cmd == "L1ON")  digitalWrite(8, HIGH);
    if (cmd == "L1OFF") digitalWrite(8, LOW);
    if (cmd == "L2ON")  digitalWrite(9, HIGH);
    if (cmd == "L2OFF") digitalWrite(9, LOW);
  }
}
```

The program reads and interprets each command to control the matching pin. Serial.begin(9600) starts serial communication at 9600 baud.

15 Explain switch bounce and the given debounce function. What change is needed for 30 ms bounce?

ANSWER

Switch bounce is the rapid unwanted ON/OFF transition of a mechanical switch. Software debouncing accepts a state only after it remains stable.

The function samples every 1 ms. If the state changes, the counter resets; after 10 stable readings, it returns the state. For up to 30 ms bounce, change `counter < 10` to `counter < 30`.

Drawback: delay() blocks the program and prevents other tasks from running during that time.

16 Identify the output: a = 15, b = 10, c = a & b.

ANSWER

15 = 1111; 10 = 1010.

1111 AND 1010 = 1010 = 10.

Output: 10

17 Identify the output: a = 40, c = a >> 3.

ANSWER

Right-shifting 40 by 3 positions is equivalent to dividing by 2^3.

40 ÷ 8 = 5.

Output: 5

18 State the logic levels of a pushbutton using the internal pull-up resistor and write its configuration.

ANSWER

With the internal pull-up resistor, the input is HIGH when the button is released and LOW when pressed.

- Pressed = LOW (0).
- Released = HIGH (1).

```
pinMode(7, INPUT_PULLUP);
```

19 Calculate the difference between two temperature sensor readings and turn ON an LED if it exceeds 15.

ANSWER

```
void setup() { pinMode(13, OUTPUT); }
void loop() {
  int s1 = analogRead(A0);
  int s2 = analogRead(A1);
  int diff = abs(s1 - s2);
  if (diff > 15) digitalWrite(13, HIGH);
  else digitalWrite(13, LOW);
}
```

The program reads both sensors, calculates their absolute difference using abs(), and turns ON the LED if the difference exceeds 15.

20 Differentiate between digital and analog sensors.

ANSWER

- **Digital sensors:** Produce discrete signals such as HIGH or LOW. Examples: PIR sensor and digital IR obstacle sensor.
- **Analog sensors:** Produce continuously varying voltage signals. Examples: LM35 temperature sensor and LDR.

21 Explain local and global variables and array indexing. Use readings 120, 135, 142 and 128; print the sum of the first and last.

ANSWER

A global variable is declared outside functions and can be accessed throughout the program. A local variable is declared inside a function or block and is accessible only there. Array indexing starts at 0.

```
int readings[4] = {120, 135, 142, 128};
void setup() {
  Serial.begin(9600);
  int sum = readings[0] + readings[3];
  Serial.println(sum);
}
void loop() { }
```

readings is a global array; sum is local. First element = 120; last = 128; output = 248.

22 Why should Serial.available() be checked before using Serial.read()?

ANSWER

Serial.available() checks whether data is present in the serial buffer. Checking it before Serial.read() ensures that valid data is available and prevents processing an empty input.

Example:

```
if (Serial.available() > 0) { char data = Serial.read(); }
```

23 Explain void setup() and void loop(). Demonstrate digitalWrite() and analogWrite().

ANSWER

- **void setup()** runs once and is used for initialization.
- **void loop()** runs repeatedly while the Arduino is powered.

```
void setup() {
  pinMode(13, OUTPUT); pinMode(9, OUTPUT);
}
void loop() {
  digitalWrite(13, HIGH);
  analogWrite(9, 128);
}
```

digitalWrite() produces HIGH/LOW output. analogWrite() produces PWM output (0–255) for controlling brightness or speed.

24 Map a potentiometer reading to LED brightness. Calculate the brightness for input 700.

ANSWER

```
void setup() { pinMode(9, OUTPUT); }
void loop() {
  int value = analogRead(A0);
  int brightness = map(value, 0, 1023, 0, 255);
  analogWrite(9, brightness);
}
```

Calculation: (700 × 255) / 1023 = 174.49 ≈ 174.

Input range = 0–1023; output range = 0–255. analogWrite() uses PWM to control LED brightness.

25 Explain bitSet(), bitClear(), bitRead(), highByte(), lowByte() and word().

ANSWER

```
byte devices = 0;
bitSet(devices, 2);
bitClear(devices, 2);
int status = bitRead(devices, 2);
unsigned int value = 258;
byte high = highByte(value);
byte low = lowByte(value);
unsigned int result = word(high, low);
```

- **bitSet()** sets a bit to 1; **bitClear()** clears it to 0; **bitRead()** reads a bit.
- **highByte()** extracts the higher byte; **lowByte()** extracts the lower byte; **word()** combines two bytes.

For 258 (0x0102): high byte = 1, low byte = 2, reconstructed value = 258.